Page replacement algorithm 24101117

1)first in first out

```c
#include <stdio.h>

int main()
{
    int i,j,n,a[50],frame[10],no,k,avail,count=0;

    printf("Enter number of pages: ");
    scanf("%d",&n);

    printf("Enter the page numbers:\n");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    printf("Enter number of frames: ");
    scanf("%d",&no);

    for(i=0;i<no;i++)
        frame[i]=-1;

    j=0;

    printf("\nRef\tFrames\n");

    for(i=0;i<n;i++)
    {
        printf("%d\t",a[i]);
        avail=0;

        for(k=0;k<no;k++)
        {
            if(frame[k]==a[i])
                avail=1;
        }

        if(avail==0)
        {
            frame[j]=a[i];
            j=(j+1)%no;
            count++;

            for(k=0;k<no;k++)
            {
                if(frame[k]!=-1)
                    printf("%d ",frame[k]);
                else
```

```c
                    printf("%d ",frame[k]);
                else
                    printf("- ");
            }
        }
        else
        {
            printf("Hit");
        }

        printf("\n");
    }

    printf("\nTotal Page Faults: %d\n",count);

    return 0;
}
```

Output:-

```
[user123@localhost bankers]$ cd page
[user123@localhost page]$ vi fifo.c
[user123@localhost page]$ gcc fifo.c
[user123@localhost page]$ ./a.out
Enter number of pages: 7
Enter the page numbers:
1 3 0 3 5 6 3
Enter number of frames: 3

Ref     Frames
1       1 - -
3       1 3 -
0       1 3 0
3       Hit
5       5 3 0
6       5 6 0
3       5 6 3

Total Page Faults: 6
[user123@localhost page]$
```

2)LRU

```c
#include <stdio.h>
int findLRU(int time[], int n)
{
    int i, minimum = time[0], pos = 0;
    for(i = 1; i < n; i++)
    {
        if(time[i] < minimum)
        {
            minimum = time[i];
            pos = i;
        }
    }
    return pos;
}
int main()
{
    int no_of_frames,no_of_pages,frames[10],pages[30],counter=0,time[10];
    int flag1,flag2,i,j,pos,faults=0;
    printf("Enter number of frames: ");
    scanf("%d",&no_of_frames);
    printf("Enter number of pages: ");
    scanf("%d",&no_of_pages);
    printf("Enter reference string: ");
    for(i=0;i<no_of_pages;i++)
        scanf("%d",&pages[i]);
    for(i=0;i<no_of_frames;i++)
        frames[i]=-1;
    printf("\nRef\tFrames\n");
    for(i=0;i<no_of_pages;i++)
    {
        flag1=flag2=0;
        for(j=0;j<no_of_frames;j++)
        {
            if(frames[j]==pages[i])
            {
                counter++;
                time[j]=counter;
                flag1=flag2=1;
                break;
            }
        }
        if(flag1==0)
        {
            for(j=0;j<no_of_frames;j++)
            {
```

```c
            if(frames[j]==pages[i])
            {
                counter++;
                time[j]=counter;
                flag1=flag2=1;
                break;
            }
        }
        if(flag1==0)
        {
            for(j=0;j<no_of_frames;j++)
            {
                if(frames[j]==-1)
                {
                    counter++;
                    faults++;
                    frames[j]=pages[i];
                    time[j]=counter;
                    flag2=1;
                    break;
                }
            }
        }
        if(flag2==0)
        {
            pos=findLRU(time,no_of_frames);
            counter++;
            faults++;
            frames[pos]=pages[i];
            time[pos]=counter;
        }
        printf("%d\t",pages[i]);
        for(j=0;j<no_of_frames;j++)
        {
            if(frames[j]!=-1)
                printf("%d ",frames[j]);
            else
                printf("- ");
        }
        printf("\n");
    }
    printf("\nTotal Page Faults: %d\n",faults);
    return 0;
}

-- INSERT --
```

output:-

```
[user123@localhost page]$ gcc lru.c
[user123@localhost page]$ ./a.out
Enter number of frames: 3
Enter number of pages: 7
Enter reference string: 7 0 1 2 0 3 0

Ref     Frames
7       7 - -
0       7 0 -
1       7 0 1
2       2 0 1
0       2 0 1
3       2 0 3
0       2 0 3

Total Page Faults: 5
[user123@localhost page]$
```